territory of the forum and the person or the activity to which the court is now interested in seeking to exercise such broader authority. Within the United States, the principle of "minimal contacts" is the doctrine of personal jurisdiction which provides that, before a "foreign" corporation is subject to suit in a "state", such foreign corporation's activity within the state must meet certain basic requirements of activity.[1] For a nonresident to be subject to a forum-state court's personal jurisdiction, he must "have certain minimum contacts with it such that the maintenance of the suit does not offend 'traditional notions of fair play and substantial justice.'"[2] Minimal contacts is a somewhat expansionistic doctrine, but it has allowed the individual states in the U.S. to respond "to the actualities of the national market"[3] by creating long-arm statutes. Long-arm statutes have limited the practice of foreign corporations avoiding legal responsibilities by scampering across state lines to avoid service of process.[4] Further, long-arm statutes provide personal jurisdiction over nonresidents of a state, when such nonresidents voluntarily enter into, or communicate with, persons in the state for limited purposes, in cases that concern claims relating to the performance or execution of such limited purposes.[5] Examples of such closely-related activities to a claim being brought forward include the transaction of business or the commission of a tortious act, either of which results in an injury to a person or damage to property within the state's borders.

So far, the primary actors in space-related activities have been governmental entities themselves. Other than privately-owned satellites and a limited handful of private citizens traveling aboard the Russian *Soyuz* spacecraft, the property of inter-

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[1] International Shoe Co. v. State of Washington, 326 U.S. 310 (1945).

[2] *Id.* at 317. (citing Milliken v. Meyer, 311 U.S. 457, 463). The Supreme Court's specific holding was that a Delaware corporation had, by its activities of employing approximately a dozen resident salesmen to sell its shoes in the State of Washington, rendered itself amenable to proceedings in the courts of that state to recover unpaid contributions to the state unemployment compensation fund exacted by the Washington Unemployment Compensation Act.

[3] Wikipedia, http://en.wikipedia.org/wiki/International_Shoe (last visited Nov. 9, 2008).

[4] *Id.*

[5] BLACK'S LAW DICTIONARY (8th ed. 2004).

est under current space-related activities is usually a space vessel or structure that is government-owned, or the space traveler is a government employee. For example, the current *International Space Station* (*ISS*) is a joint project between five governmentally-supported space agencies: the National Aeronautics and Space Administration (NASA, United States), the Russian Federal Space Agency (Roskosmos, Russian Federation), the Japan Aerospace Exploration Agency (JAXA, Japan), the Canadian Space Agency (CSA, Canada) and the European Space Agency (ESA, Europe), which is in turn comprised of eleven member nations.

The growth of government participation by other nations in space-related activities has been further encouraged by President George W. Bush in his 2004 Vision for Space Exploration, in which he announced a plan to return to the Moon by the year 2020. President Bush directed NASA to build "on its long history and extensive and close ties with the space and research agencies of other nations... [by] seek[ing] international partners and work[ing] with the space agencies of these partners in executing future exploration activities."[6] To achieve this goal, NASA's Exploration System Mission Directorate, charged with the responsibility of developing future human exploration programs, has been pursuing broad international partnerships with other nations for a future lunar base. Thus, at least from the perspective of governmental-based activities, it appears that multi-national partnerships for space-related activities and the proximal interconnectivity of multinational spacecraft that result from these partnerships will continue to exist and should grow.

Government participation in the space-related activity of interest would seemingly provide enough of a nexus to support a claim of jurisdictional interest by a court of corresponding national origin. Yet private commercial activities are on the rise, and economic growth in this area can be expected to continue in the coming years. It is even conceivable that human presence in

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[6] NATIONAL AERONAUTICS AND SPACE ADMINISTRATION, THE VISION FOR SPACE EXPLORATION 17 (Feb. 2004), *available at* http://history.nasa.gov/Vision_For_Space_Exploration.pdf.

low-earth orbit funded entirely by a private commercial entity will occur by the beginning or middle of the next decade. Would a national government's participation in the development of a multi-national spacecraft also provide enough of a nexus to support a claim of jurisdiction to activities occurring therein by the corresponding national court?

For instance, the *ISS* has been designed for conducting scientific research and experimentation unique to a microgravity environment by highly skilled astronauts and scientists. Experiments to be carried out on the *ISS* include the scientific fields of human physiology, medicine, biology, propulsion systems, and long-term space exposure (just to name a few). In particular, the pharmaceutical sector has been identified as a scientific field that will, in all likelihood, benefit from experiments carried out on the *ISS*. Currently, the United States (U.S.), ESA, and Japan have on-orbit, operating, ISS-laboratory modules specifically designed for conducting such experiments. The Russian Federation has plans for its own laboratory module with an estimated launch date of December 2011. In terms of the future status of the U.S. segments on the *ISS*, once *ISS* assembly is complete, NASA plans to transition the U.S. segments of the *ISS* into a National Laboratory and dramatically increase commercially-funded research and development activities onboard the *ISS*. While still in the planning stages, a future National Laboratory is currently envisioned to continue to operate within the framework of the *ISS* and the existing international agreements. The *ISS* thus represents a novel "territory" for potential innovations that may result in patentable inventions as well as a resource wherein patented inventions are used or practiced. One may reasonably expect patentable inventions to be conceived on the *ISS*. In addition, patentable inventions may be conceived on Earth and actually reduced to practice or used for their intended purposes on the *ISS*.

The question of extra-territorial reach of one nation's courts to activities, persons, and structures in outer space presents an opportunity to re-examine the established notion of restraint when exercising jurisdiction beyond traditional territorial borders. Outer space, including celestial bodies and the Moon, are territories that are considered to be, at least initially, *"res com-*

munis", that is, things common to all and that cannot be owned or appropriated.[7] The lack of true territorial jurisdiction in space poses some interesting questions to consider: What constitutes an appropriate expansion of jurisdiction by a nation and its courts into a territory that cannot be owned or appropriated by any nation? Should some type of nexus exist between a nation and the person or the activity of jurisdictional interest, as is currently required for personal jurisdiction within the United States under the principle of minimal contacts and its related long-arm statutes, if such nation seeks to exercise a broader authority over space-related activities? Will an international treaty, an agreement among a group of space-faring nations that includes the United States, or a legislative act of Congress suffice to bring appropriate authority for a jurisdictional claim when private corporate entities of other third-party nations might emerge as a primary participant in human space travel?

An attempt to address all of these questions is beyond the scope of this article. Rather, this article will focus on issues related to the expansion of territorial jurisdiction for the protection of intellectual property associated with space-related activities. Specifically, we will examine this question: "What laws, if any, and in particular, what U.S. patent laws, apply to inventions and patent rights for activities in outer space?" U.S. patent law has several specific statutory provisions that concern jurisdictional and extra-territorial reach.[8] We will examine a number of these statutory provisions and the particularities they present. As an outline, this article will address the following areas:

- brief overview of patents and the U.S. Patent system;
- introduction to the Outer Space Treaty and the ISS Intergovernmental Agreement;
- extra-territorial reach as it relates to infringement under U.S. patent law;

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[7] BLACK'S LAW DICTIONARY, *supra* note 5.

[8] *See e.g.*, 35 U.S.C. §§ 102(a), (b), (g)(2), 104(a), 105, 271-272.

- the Temporary Presence Doctrine;
- identification of potential issues by analysis of exemplary hypothetical cases; and
- possible techniques for avoiding or mitigating jurisdictional problems as they relate to patent law.

The article's case analysis of the hypothetical cases will seek to assess potential patent infringement issues on the *ISS* and at a future international lunar base resulting from the close proximity of habitable modules registered under a variety of nations. Further, commercial scenarios in outer space will be evaluated for potential patent infringement issues.

II. BRIEF OVERVIEW OF PATENTS

In most nations, a patent represents a property right granted by the national government for a fixed period of time to an applicant who is an inventor or group of inventors, or to a representative thereof, which made a new invention. This property right is normally limited to the territorial reach of the granting nation.[9] Hence, patent law is inherently territorial in nature. Once issued, a patent authorizes its owner(s) the right to exclude others from practicing or otherwise engaging in a particular activity.[10] The grant and enforcement of patents are based on and governed by national laws, or on occasion by international treaties that have given regional effect to a nationally issued patent.

The scope of an invention is defined by the claims of a patent. A "claim" is a legal description of the subject matter that an inventor regards as his invention.[11] The property right associated with the patent allows a patent owner the right to exclude others from practicing the claimed invention. Patents may be sold, purchased, licensed, or otherwise exchanged in a

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[9] *See e.g.,* Deepsouth Packing Co. v. The Laitram Corporation, 406 U.S. 518, 532 (1972) (superseded on other grounds by Patent Amendments Act of 1984, Pub. L. No. 98-622, 98 Stat. 3383 (now codified at 35 U.S.C. §§ 271(f))) [hereinafter *Deepsouth*].

[10] *See* 35 U.S.C. § 154.

[11] *See* 35 U.S.C. § 112.

manner similar to real or personal property. The right to exclude others from practicing the claimed invention does not necessarily give the patent owner the right to practice the claimed invention himself. Instead, in order to practice the claimed invention, the patent owner may have to comply with other laws or regulations to practice the invention, obtain licenses from other patent owners, or any combination thereof.

To obtain a patent, an inventor or group of inventors must first file a patent application [hereinafter application] in a jurisdiction of interest. The inventor or group of inventors who file an application is referred to as the Applicant or Applicants, respectively. An application may begin as an international or "PCT" (Patent Cooperation Treaty) application, which enables a domestic application to be later filed in a particular country or group of countries. However, an application under the PCT is incapable of enforcement unless the application enters the national phase by a subsequent filing of an application in an appropriate governmental office responsible for processing the application for a particular "State" or "Region."

In the U.S., the applicable governmental office is the United States Patent and Trademark Office or "USPTO." The USPTO employs a number of scientists and engineers as "Examiners," who analyze or examine applications to determine patentability. Examiners, skilled in their particular area of technology, are charged with analyzing the application in light of the "prior art" and patent laws to determine if the application should be permitted to issue as a patent. If the Examiner determines that any claim of the patent application does not meet the statutory criteria for a patent, the Examiner will reject the claim. The Examiner's basis for any rejection is documented in a communication known as an "Office Action," to which the Applicant, typically represented by a patent attorney or agent, responds. An Office Action may contain more than one ground for rejection, each of which must be addressed in the Applicant's response. Currently, the examination process in the U.S. generally takes about three years from the filing date of a final or "non-provisional" application. If all grounds for rejection have been adequately addressed by the Applicant to the Examiner's satisfaction, the application will ultimately issue as a patent.

III. INTRODUCTION TO THE OUTER SPACE TREATY AND THE ISS INTERGOVERNMENTAL AGREEMENT

Perhaps the formative instrument related to establishing an international legal framework for activities conducted in outer space is the 1967 "Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space Including the Moon and Other Celestial Bodies,"[12] [hereinafter Outer Space Treaty]. All major space-faring nations have signed and ratified the Outer Space Treaty. Under this treaty, the signatories agreed, "A State Party to the Treaty on whose registry an object launched into outer space is carried shall retain jurisdiction and control over such object, and over any personnel thereof, while in outer space or on a celestial body."[13] This principle is analogous to the "floating island" principle existing in maritime law for ships in international waters. Further, the signatories agreed, "Outer space, including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means."[14] This principle, described above as *res communis*, has been negotiated and accepted as terms for international treaties concerning terrestrial geographic areas, such as Antarctica and international waters. Such territories do *not* fall under any state's territorial jurisdiction.[15]

In January 1998, fifteen nations executed an international treaty know as the "ISS Intergovernmental Agreement" or "IGA." The IGA provides a "framework on the basis of genuine partnership, for the detailed design, development, operation, and utilisation of a permanently inhabited civil Space Station for peaceful purposes, in accordance with international law."[16]

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[12] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, Oct. 10, 1967, 18 U.S.T. 2410, T.I.A.S. 6347 [hereinafter Outer Space Treaty].

[13] *Id.* at art. VIII.

[14] *Id.* at art. II.

[15] *See e.g.*, Smith v. United States, 507 U.S. 197 (1993) (holding that the Federal Tort Claims Act does not apply to tortuous acts or omissions occurring in Antarctica, a sovereignless region without civil tort law of its own).

[16] Agreement Among the Government of Canada, Governments of the Member States of the European Space Agency, the Government of Japan, the Government of the

Article 21 of the IGA specifically addresses Intellectual Property. Article 21 states,

> for the purposes of intellectual property law, any activity occurring in or on a Space Station flight element shall be deemed to have occurred only in the territory of the Partner State of that element's registry, except that that for ESA-registered elements any of the eleven European Partner States may deem the activity to have occurred within its territory.[17]

Thus, the intellectual property provisions of the IGA followed the general principle of national jurisdiction found in the Outer Space Treaty for activities, including inventions occurring within registered space objects. For example, an invention made on a Russian Federation segment of the *ISS* will be deemed to have occurred in Russia, and Russian patent law will apply. The situation becomes somewhat more complicated if an invention is made on an ESA-registered segment. Specifically, any European Partner State may elect to deem the inventive activity occurred within its territory. The compli-cations of this arrangement of concurrent jurisdiction will shortly be examined and addressed below.

Article 21 also addresses the Partner States' respective invention secrecy laws. In the U.S., the Invention Secrecy Act restricts the filing or dissemination of patent applications for national security purposes. Similarly, the national laws of many other states impose a delay or require prior authorization before a patent application can be filed in a foreign country. In the U.S., a foreign filing license[18] is required from the USPTO before an application may be filed in a foreign country. Article 21 of the IGA places some limitation on the application of a Partner State's invention secrecy laws.[19] In general, for an invention made on the *ISS*, one must: (1) determine where the invention was "made" (i.e., which country registered the specific

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Russian Federation, and the Government of the United States of America Concerning Cooperation on the Civil International Space Station, Jan. 28, 1998, at art. 1 [hereinafter IGA].

[17] *Id. at* art. 21.2.

[18] 35 U.S.C. § 184.

[19] IGA, *supra* note 16, at art. 21.3.

ISS module); (2) determine the nationality of the inventor; and (3) determine if the invention rises to the level of being designated as embodying classified information or otherwise protectable under the concept of "national security." As an example, if the invention was made by a Canadian national in the U.S. Lab module of the *ISS* and is considered a national security asset, article 21 states that the U.S. cannot impose its invention secrecy laws and restrict the Canadian national inventor from filing a patent application in Canada (provided Canada has invention secrecy laws). Once an individual has filed a patent application in a particular Partner State, that State may subsequently apply its invention secrecy laws.

Article 21 also addresses infringement and licensing issues associated with the concurrent-jurisdictional arrangement for the member States of the ESA. For example, consider the scenario wherein an invention, patented in all ESA member States, is used or performed onboard the ESA Lab (*Columbus*) module. Under the concurrent-jurisdictional arrangement, the patent owner could theoretically file a patent infringement suit in all eleven ESA member States for a single infringing act. Article 21 remedies this potential outcome by stating that "where a person or entity owns intellectual property which is protected in more than one European Partner State, that person or entity may not recover in more than one such State for the same act of infringement."[20] Article 21 also provides that "satisfaction of a judgment rendered for damages...shall bar further recovery of damages in any pending or future action for infringement based upon the same act of infringement." In terms of licensing issues, article 21 states that "no European Partner State shall refuse to recognize a license for the exercise of any intellectual property right if that license is enforceable under the laws of any European Partner State." Thus, if a licensee has the right to practice a patented invention in any European Partner State, that licensee has the right to practice the patented invention in or on an *ISS* ESA module.

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[20] *Id.* at art. 21.4.

In sum, both the Outer Space Treaty and the IGA have attempted to address territorial jurisdictional questions, even issues associated with intellectual property. While this international treaty and agreement, respectively, have been made among space-faring nations that include the United States, they leave open the question as to what would be the appropriate authority for a jurisdictional claim when private corporate entities are the primary actors in space-related activities.

IV. EXTRA-TERRITORIAL REACH AS IT RELATES TO INFRINGEMENT UNDER U.S. PATENT LAW

As mentioned earlier in Part II, inventors may obtain a patent from the USPTO if their invention is determined to be useful[21], novel[22], nonobvious[23], sufficiently described and enabled in the patent application.[24] An inventor who has obtained a patent from the USPTO is generally referred to as a "Patentee." In general, a Patentee can sue in a U.S. Federal Court if there has been an unauthorized, statutorily-defined activity related to the Patentee's claimed invention. Such unauthorized, statutorily-defined activities are given in Title 35 of the United States Code (U.S.C.), Chapter 28, which forms the foundation of what constitutes a "patent infringement" in the United States. The core of the U.S. Patent Act's provisions on infringement has historically been explicitly territorial in its jurisdictional scope of infringing subject matter. For example, § 271(a), as currently amended, states:

> Except as otherwise provided in this title, whoever without authority makes, uses, offers to sell, or sells any patented invention, *within the United States*, or imports *into the United States* any patented invention during the term of the patent therefor, infringes the patent." (emphasis added)

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[21] 35 U.S.C. § 101 states, "Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefore, subject to the conditions and requirements of [U.S. patent law]."

[22] 35 U.S.C. § 102.

[23] 35 U.S.C. § 103.

[24] 35 U.S.C. § 112.

We will now discuss some court decisions that have examined the territorial scope of the U.S. Patent laws, both before and after the passage of the modern-day patent act,[25] in association with different types of actors before the court and with varying degrees of activity occurring beyond the borders of the United States. Discussion of these cases provides insight into the framework of a court's analysis of the jurisdictional issue, which generally favors a presumption against extra-territorial reach. The U.S. courts have traditionally focused carefully on the specific facts presented to determine whether the subject matter of each case, examining both the actor and the activity of concern, provides an appropriate nexus to hold that the activity is within its territorial reach. These cases thereby shed light on what space-based activities might also be found to be "within the United States" for issues related to patent infringement.

A. Territorial Scope before the U.S. Patent Act of 1952

Even before the enactment of the U.S. Patent Act of 1952, the U.S. Supreme Court had taken a restricted view of subject matter jurisdiction with respect to the territorial reach of U.S. patent law, favoring a presumption against an extra-territorial effect.[26] For instance, in 1856 the Supreme Court refused to extend the rights of an owner of a U.S. patent to apply to the activities of a French schooner that had temporarily docked at a Boston port.[27] The defendant in *Brown*, who was the captain of the schooner, had answered the complaint of the patent owner by demurrer and pleaded that he and his ship were subjects of the French empire, that he was temporarily located in the United States, and that the alleged infringement occurred while

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[25] U.S. Patent Act of 1952, 35 U.S.C. §§ 1-376, Pub. L. No. 593 (July 19, 1952) (including amendments).

[26] See Robert W. Pierson, *Extraterritorial Reach of U.S. Patent Law: Has the Federal Circuit Gone Too Far?*, 17 FORDHAM INTELL. PROP. MEDIA & ENT. L.J. 651, 655 (2007), *available at* http://law.fordham.edu/publications/articles/200flspub7855.pdf (last visited on Nov. 9, 2008).

[27] Brown v. Duchesne, 60 U.S. 183 (1856) (as support for its ruling, the court opined "acts of Congress do not and were not intended to operate beyond the limits of the United States, and as the patentee's right of property and exclusive use is derived from them, they cannot extend beyond the limits to which the law itself is confined.").

upon a lawful voyage between a French colony and Boston.[28] The Court framed the issue to be considered as "whether any improvement in the construction or equipment of a foreign vessel for which a patent has been obtained in the United States can be used by such vessel within the jurisdiction of the United States while she is temporarily there for the purposes of commerce, without the consent of the patentee?"[29] Thus, the Court was presented with a mixed question of personal and subject matter jurisdiction in that both the infringing actor and the infringing activity had a temporary connection to the United States. The Court, however, apparently considered the territorial reach at issue in this case of infringement to rest more squarely on the shoulders of subject matter rather than personal jurisdiction. Specifically, the Court stated:

> This question depends on the construction of the patent laws. For undoubtedly every person who is found within the limits of a government, whether for temporary purposes or as a resident, is bound by its laws. The doctrine upon this subject is correctly stated by Mr. Justice Story in his "Commentaries on the Conflict of Laws" chap. 14, sec. 541, and the writers on public law to whom he refers. A difficulty may sometimes arise, in determining whether a particular law applies to the citizen of a foreign country and intended to subject him to its provisions. But if the law applies to him and embraces his case, it is unquestionably binding upon him when he is within the jurisdiction of the United States.[30]

The *Brown* Court found that the infringing equipment had been placed upon the vessel before entry into the United States.[31] The Court also determined that the equipment was not for use while "at anchor in the port" and that the equipment had not been "manufactured on her deck while she was lying in the port of Boston."[32] Given that the alleged infringing activity relating to the foreign vessel had occurred while away from port,

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[28] *Id.* at 193.
[29] *Id.* at 194.
[30] *Id.* at 194.
[31] *Id.* at 195-96.
[32] *Id.* at 196.

the Court concluded that the defendant's use "was on the high seas and in other places out of the jurisdiction of the United States."[33]

Similarly, in 1915, the U.S. Supreme Court, while considering a patentee's rights to recover damages related to a patented component known as "shoe drills", denied recovery for damages attributable to the defendant's sales of the infringing product in Canada, even though the grain drill incorporating the patented component had been made and purchased by the defendant in the United States.[34] The Court noted that a third party had made the infringing drills in the United States and then sold them to the defendant.[35] In finding no liability for the sales activity of the defendant that took place in Canada, the Court explained that "[t]he right conferred by a patent under our law is confined to the United States and its territories, and infringement of this right cannot be predicated of acts wholly done in a foreign country."[36] The Court determined that the alleged infringing activities of the defendant were beyond the territorial reach because the place of sale was "of controlling importance here."[37]

The first reported expansion of the "territorial" reach of U.S. patent law appears to have occurred in a somewhat obscure 19th Century decision out of the Circuit Court of Massachusetts that concerned an infringement claim against an American vessel.[38] The facts in *Gardiner* were somewhat similar to the facts of *Brown*. The plaintiff sought to recover damages for the infringement of a patent to the improvement of a sail for vessels, except in this case the vessel was American and commanded by an American "master" on her passage from Liverpool to New York.[39] The defendant sought to defend the claim by showing that the patented invention was used "on the high

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[33] *Id.* at 196.
[34] Dowagiac Mfg. Co. v. Minnesota Moline Plow Co., 235 U.S. 641 (1915)
[35] *Id.* at 650.
[36] *Id.* at 650.
[37] *Id.* at 650.
[38] Gardiner v. Howe, 2 Cliff. 462, 9 F.Cas 1157 (C.C.D. Mass. 1865) (No. 5219).
[39] *Id.*

seas, and without the jurisdiction of the United States"[40], citing as authority the U.S. Supreme Court's decision of *Brown v. Duchesne*.[41] The *Gardiner* Court found that *Brown* could be readily distinguished here in that the vessel at issue was American. Specifically, the *Gardiner* Court held that while "[t]he patent laws of the United States afford no protection to inventions beyond or outside of the jurisdiction of the United States[,]...this jurisdiction extends to the decks of American vessels on the high seas, as much as it does to all the territory of the country, and for many purposes is even more exclusive."[42] The *Gardiner* Court reasoned that "[w]ere it to be held that in cases like the present the plaintiff is not entitled to recover, patents for improvements in the tackle and machinery of vessels, or in their construction, would be valueless."[43] In one sense, the *Gardiner* Court's rationale for expanding its territorial reach appears to rely on its own notion of fairness, a rationale that has since evolved to find support in the jurisprudence of personal jurisdiction through the principle of minimal contact, i.e., "fair play and substantive justice." This notion of fairness, however, has not expressly been relied on by the courts in determining subject matter jurisdiction under the U.S. Patent laws.

The jurisdictional theory of *Gardiner* also differs from the jurisdictional approach of *Brown* in that it is based on the nationality of the vessel as opposed to the geographical location of the vessel. National jurisdiction is "based on the nationality of persons or things subject to state control."[44] Conversely, territorial jurisdiction is based on the actual borders defining the geographic limitations of a nation. These two principles "are often confused" with one another "but they are in fact distinct".[45] Given that Congress has subsequently amended the Patent Act

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[40] *Id.*

[41] *Brown, supra* notes 27 - 33.

[42] *Gardiner, supra* note 38, at 462.

[43] *Id.*

[44] See Glenn H. Reynolds, *Legislative Comment: The Patents in Space Act*, 3 HARVARD J.L. & TECH. 13, 18 (1990), *available at* http://jolt.law.harvard.edu/articles/pdf/v03/03HarvJLTech001.pdf (last visited on Nov. 9, 2008).

[45] *Id.*

to define the United States as being limited outside its borders to "its territories and possessions", substantive arguments can be readily made that the outcome in *Gardiner* was incorrect in finding territorial reach under U.S. patent law for activity of a private entity, even though U.S. flagged, on the high seas.[46] One commentator has even suggested that a possible purpose of Congress in adding this language to the patent law was specifically to overrule *Gardiner*.[47] Nevertheless, the extra-territorial reach found by the *Gardiner* Court, applying U.S. law to U.S. flagged vessels on the high seas, has been embraced and adopted as the jurisdictional principle of both the Outer Space Treaty and the IGA.

B. Territorial Scope After the U.S. Patent Act of 1952

Congress's passage of the U.S. Patent Act of 1952 [hereinafter *Patent Act*] seemingly endorsed the Court's longstanding tradition of the limited territorial scope of U.S. patent law. For instance, the Act included a provision entitled "Temporary Presence in the United States",[48] which provided that "[t]he use of any vessel aircraft or vehicle of any country" that affords similar privileges to those of the United States shall not constitute infringement when entering the United States temporarily or accidentally, "if the invention is used exclusively for the needs of the vessel, aircraft or vehicle and is not sold in or used for the manufacture of anything to be sold in or exported from the United States."[49] This statute was a codification of the Supreme Court's *Brown* decision. The Patent Act also included a section entitled "Definitions" that provided that "[t]he terms 'United States' and 'this country' mean the United States of

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[46] See for instance the Supreme Court's discussion of this Congressional act in *Deepsouth, supra* note 9, at 530.

[47] Reynolds, *supra* note 44, at 21. The commentator opens his criticism of the *Gardiner* decision (at p. 20) by stating that "case law purporting to support the application of U.S. patent law to U.S. space objects is of doubtful value", but the commentator also notes at p. 21 that the outcome of *Gardiner* "was correct on policy if not legal grounds." *Id.*

[48] Codified at 35 U.S.C. §§ 272.

[49] U.S Patent Act of 1952, *supra* note 25, at ch. 950.

America, its territories, and possessions."[50] Finding little to dissuade its limited view of territorial scope of U.S Patent law, the U.S. Supreme Court continued to favor a presumption against extraterritorial effect unless there was "a clear congressional indication of intent to extend the patent privilege."[51] Congress eventually accepted the Court's invitation in *Deepsouth* to provide such clear intent, enacting § 271(f) of Title 35, in essence creating a statutory exception to the Court's *Deepsouth* holding by making it an infringement to export unpatented components of a patented combination invention for later completed assembly abroad.[52] As will be discussed *infra*, Congress has also sought to expand the territorial reach of Title 35 to space objects and components.[53]

The expansion of extra-territorial reach of U.S. patent law in the context of U.S. citizens and spacecraft on the Moon was addressed by the USPTO's Board of Appeals (the predecessor to the Patent Office Board of Appeals and Interferences) in *Ex Parte McKay*.[54] In this case, the USPTO Board of Appeals was concerned with an earlier rejection of a patent application for a process of obtaining oxygen from extra-terrestrial materials in one representative set of the claims and from lunar surface material in another representative set of the claims. The patent application's specification described the use of the process on the Moon.[55] The Examiner's rejection was based partially on a strict territorial argument that U.S. patent law does not extend to inventions whose "process is to be carried out outside the United States"[56] (in this case the Moon), citing 35 U.S.C. § 100 and § 154. While rejecting the application under other grounds of obviousness, the Board of Appeals rejected the Examiner's territorially-based rejection. The Board first recognized that the *res communis* principle of article II of the Outer Space Treaty was somewhat at odds with the subject matter of the claims of

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[50] *Id.* at § 100(c).
[51] *See, e.g., Deepsouth supra* note 9, at 532 (1972).
[52] Patent Amendments Act of 1984, *supra* note 9.
[53] 35 U.S.C. § 105.
[54] *Ex Parte McKay*, 200 U.S.P.Q. 324 (Pat.Off.Bd.App. 1975).
[55] *Id.* at 325
[56] *Id.*

the patent application and the extension of jurisdiction by means of the national jurisdictional principle of article VIII. Ultimately relying on article VIII of the treaty to find jurisdiction, the Board of Appeals held,

> [i]t is clear from article VIII of said Treaty that jurisdiction of the United States in personam over any person is present if the object launched into outer space is of United States registry. A patent grant under 35 U.S.C. § 154 by the United States for a process to be carried out on the moon by personnel subject to its jurisdiction is thus not inimical and at variance with

U.S. patent law.[57] Thus, the Board of Appeals relied on the policies defined in the Outer Space Treaty as opposed to the law as defined in Title 35 of the United States Code. Given that the United States has been clearly defined in the Patent Act to be limited outside its borders to "its territories and possessions,"[58] commentators have suggested that the reasoning in *Ex Parte McKay* was incorrectly decided in finding *in personam* jurisdiction.[59] Thus, even after *Ex Parte McKay*, questions remained as to the applicability of Title 35 to activities aboard vehicles in outer space due to the lack of explicit territorial language in the U.S. patent law relating to outer space.

Another case related to territorial scope after the Patent Act is *Decca Limited v. United States*.[60] Decca brought a 28 U.S.C. § 1498 action against the U.S. Government for patent infringement. Although technically not a 35 U.S.C. § 271(a) case, the territorial requirements of what constitutes an infringement by the U.S. Government as compared to a non-governmental entity are essentially the same. The alleged infringing technology involved a worldwide radio navigation system with two stations in the United States and one station in Norway. The subject system was designed and built by the U.S. Government. Further, the technology was dependent on at least

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[57] *Id.*

[58] 35 U.S.C. § 100(c).

[59] *See* Reynolds, *supra* note 44.

[60] Decca Limited v. United States, 210 Ct. Cl. 546, 552-53, 544 F.2d 1070, 1074-75 (Ct. Cl. 1976) [hereinafter *Decca*].

two stations, one of which served as a "master station," with the second station being controlled by the master station to ensure signal synchronization. A third station was required for position location and was also controlled by the master station. In practice, the actual master station was located in the U.S. Ultimately, the Court found, "[1] ownership of the equipment by the United States, [2] the control of the equipment from the United States and ... [3] the actual beneficial use of the system within the United States"[61] justified the application of U.S. law. Thus, in the absence of unauthorized infringing activities entirely within the territorial boundaries of the U.S., the Court seemingly created a three-prong test for finding such activity "within the United States": (1) U.S. ownership, (2) control from the U.S., *and* (3) beneficial use within the U.S.

The extra-territorial reach of U.S. patent law to a transnational telecommunications system was re-examined in *NTP v. Research in Motion*.[62] This case, in part, directly addressed 35 U.S.C. § 271. The subject technology related to systems for integrating existing electronic mail systems ("wireline" systems) with radio frequency (RF) wireless communication networks, to enable a mobile user to receive email over a wireless network.[63] The technology was integrated in the Canadian-based Research in Motion's BlackBerry® handheld devices, which provided mobile email access to users of such devices in the United States by use of Research in Motion's wireless network located in Canada. As to the issue of whether or not the BlackBerry® devices infringed NTP's patents, the Court considered the question of "whether the using, offering to sell, or selling of a patented invention is an infringement under section 271(a) if a component or step of the patented invention is located or performed abroad."[64] NTP's patented technology consisted of system (i.e., apparatus) and method (i.e., process) claims. The Court addressed what constitutes "use" in the context of both a system

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[61] *Id.* at 1083.

[62] NTP, Inc. v. Research in Motion, Ltd., 418 F.3d 1282 (Fed. Cir. 2005) [hereinafter *NTP*].

[63] *Id.* at 1287.

[64] *Id.* at 1315.

and a method claim. The Court cited *Decca* as instructive[65] and providing a "legal framework"[66] for analyzing its case. Yet, in regard to the system claim, the Court stated, "[t]he use of a claimed system under section 271(a) is the place at which the system *as a whole* is put into service."[67] The Court used two of the *Decca* prongs for its consideration. Specifically, "control of the system" and "beneficial use" were cited by the Court.[68] "Ownership," on the other hand, was not used by the Court in its analysis.[69] As to the method claims, the Court used a different approach to determine whether there was an infringement "within the United States." The Court stated, "[w]e therefore hold that a process cannot be used 'within' the United States as required by section 271(a) unless each of the steps is performed in [the United States]."[70] Thus, the Court applied different tests for determining territorial reach for system and method claims directed to essentially the same technology. For system claims, the Court decided to use two of the *Decca* prongs ("control" and "beneficial use") but not the third ("ownership") while presenting a strict territorial test for the place of performing acts of the method claims.

Although these cases provide, as the *NTP* Court states, a legal framework for addressing the issue of what constitutes "within the United States" for purposes of patent infringement, the only consensus that these cases seemingly offer is that the ultimate determination of what constitutes an unauthorized activity "within the United States" is a mixed question of fact and law to be determined by the subject matter, in particular the location and extent of the allegedly infringing activity and the type of claims in the patent, on a case-by-case basis. As one commentator has noted, "[c]ourts have provided extraterritorial effect to U.S. patents not by broadly interpreting the Patent Act,

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[65] *Id.*

[66] *Id.* at 1316.

[67] *Id.* (emphasis added).

[68] *Id.* at 1317.

[69] Research in Motion is a Canadian corporation. If the Court had used all three "*Decca*" prongs, it would have reached a different holding because there was no U.S. "ownership" in regards to Research in Motion.

[70] *NTP, supra* note 62, at 1318.

but rather by broadening the concept of subject matter considered to be inside the United States."[71]

C. Extra-Territorial Reach Under the Inventions in Outer Space Act of 1990

Jurisdictional questions related to space-based activities were to a certain extent resolved in 1990 when Congress added 35 U.S.C. § 105, entitled "Inventions in Outer Space" to U.S. patent law. 35 U.S.C. § 105 recites its general rule in the beginning of paragraph (a): "Any invention *made, used, or sold* in outer space on a space object or component thereof under the jurisdiction or control of the United States shall be considered made, used or sold within the United States for the purposes of this title...."[72]

The statute continues by providing exceptions to this general rule. The statutory exceptions include "any space object or component thereof that is specifically identified and otherwise provided for by an international agreement to which the United States is a party."[73] Further, 35 U.S.C. § 105(b) includes an exception whereby an invention "made, used, or sold in outer space on a space object or component thereof" registered in a foreign country can be considered to be "made used or sold within the U.S." if agreed upon in an international agreement between the U.S. and the state of registry.[74] Another statutory exception in 35 U.S.C. § 105(a) provides that even if a space object or component thereof is under the jurisdiction or control of the U.S., it will *not* be considered U.S. territory if it is registered by a foreign state in accordance with the Registration of Objects Launched into Outer Space [hereinafter Registration Convention]. Thus, an international agreement to which the U.S. is a

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[71] *See* Pierson, *supra* note 26, at 655.

[72] 35 U.S.C. §§ 105(a) (emphasis added).

[73] *Id.*

[74] "Any invention made, used, or sold in outer space on a space object or component thereof that is carried on the registry of a foreign state in accordance with the Convention on Registration of Objects Launched into Outer Space, shall be considered to be made, used, or sold in the United States for the purposes of this title if specifically agreed in an international agreement between the United States and the state of registry." 35 U.S.C. §105(b).

party and the registration of a space object or component thereof can effectively define the territorial jurisdiction of any patented inventions made, used, or sold in outer space on the space object or component thereof.

The language "jurisdiction or control of the United States" in 35 U.S.C. § 105(a) presents some interesting fodder for discussion and debate. The statute's legislative history[75] explains that the term "under the jurisdiction *or* control" (emphasis added) was modified[76] from article VIII of the Outer Space Treaty,[77] which uses the term "jurisdiction *and* control." One could argue that because this term originated from article VIII of the Outer Space Treaty, it is more likely than not that the word "control" is related to "jurisdiction" and thereby effectively means "jurisdictional control" as opposed to a connotation of "operational control." Conversely, one could counter that because the conjunction was changed from "and" to "or," the drafters of § 105 intended a different meaning for the word "control" and its related phrase of "jurisdiction or control" as compared to the Outer Space Treaty. This distinction is important because if one argues the latter interpretation, the object of the preposition (i.e., United States) would support an interpretation that means the control of the United States Federal Government, such as NASA or the Department of Defense. Conversely, if one argues the former interpretation, the "United States" would likely mean the United States as a geographical territory embodied by its national government, thereby invoking the jurisdiction of the U.S. courts. Both of these interpretations of the term "United States" appear in different parts of 35 U.S.C. § 105.[78] The legis-

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[75] S. REP. NO. 266, 101st Cong. (1990) at 7.

[76] *Id.*

[77] "A State Party to the Treaty on whose registry an object launched into outer space is carried shall retain jurisdiction and control over such object, and over any personnel thereof, while in outer space or on a celestial body", Outer Space Treaty, *supra* note 12, at art. VIII (emphasis added).

[78] Consider the language, "made, used, or sold in the United States." The "United States" in this context means, arguably, a national state entity geographically defined by territorial borders. Further, consider the language, "to which the United States is a party." The "United States" in this context means, arguably, the U.S. Federal Government. Further, the legislative history of § 105 specifically states that "to which the United States is a party" encompasses a Federal agency of the U.S. Government (*see* S. REP. NO. 101-266, at 7).

lative history notes that U.S.-registration of a space object in accordance with the Registration Convention is to be taken as determinative of U.S. jurisdiction and control, absent an exception defined in § 105(a).[79] Absent an actual registration, the legislative history of 35 U.S.C. § 105 is relatively silent on what constitutes "jurisdiction or control" of the United States. Hence, it remains to be seen how this language will be interpreted by U.S. Courts.

Other phrases appearing in the language of 35 U.S.C. § 105 present some further areas ripe for discussion and debate. For example, 35 U.S.C. § 105 and its legislative history does not adequately address what is meant by the term "components" of a space object. The legislative history does state that "space object" is more expansive as compared to a "space vehicle" and "is used as defined in the [Registration Convention]."[80] Thus, it appears that the term "space object" was used to encompass satellites and the like as compared to being limited to human-rated transportation or supply vehicles. But, does the term "components" encompass integrated components of a space object; stand-alone components necessary for the operation and exclusive needs of a space object; any components carried on the space object (e.g., logistical supplies or stand-alone test devices); or all of the above? This question is not addressed in the legislative history of 35 U.S.C. § 105 and could form the basis for future litigation.

One of the few cases to examine the Inventions in Outer Space Act after its passage of 35 U.S.C. § 105 is *Hughes Aircraft Co. v. United States*.[81] While determining that the Act had no "direct effect" to the outcome of the case because the launching activity had occurred before the Act's passage, the Court noted

> [t]he legislative history indicates that the purpose of the law was 'to clarify U.S. patent law with respect to its extraterritorial application aboard U.S.-flag spacecraft, in order to encourage private investment in research and manufacture con-

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[79] S. REP. NO. 266, 101st Cong. (1990) at 7.

[80] *Id.*

[81] Hughes Aircraft Co. v. United States, 29 Fed. Cl. 197 (1993) [hereinafter *Hughes*].

> ducted in outer space.'... Moreover, the legislative history suggests that the Act was consistent with international law.[82]

After commenting on the inapplicability of 35 U.S.C. § 105, the Court in *Hughes* proceeded to address the issue of what constitutes "within the United States" as applied to technology designed for use on a satellite orbiting the Earth. *Hughes* owned a patent that claimed an apparatus for controlling the attitude of a spin-stabilized spacecraft. *Hughes* brought a 28 U.S.C. § 1498 action against the United States Government seeking compensation for the alleged use or manufacture by the United States Government "of spacecraft containing an embodiment of the patented apparatus." In its analysis of what constitutes "within the United States," the Court provided an interesting analysis relating to one particular satellite, *ARIEL 5*. *ARIEL 5* never entered the U.S.; was built in the United Kingdom (U.K.); was launched off the coast of Kenya by a team of Italian engineers from a U.S.-provided launch vehicle; and was primarily controlled from the U.K.[83] Absent a physical presence in the United States, the Court looked to whether there existed a *direct* control point in the United States. Tracking and data relay services were controlled from the U.S., but core satellite system monitoring and control were performed in the U.K. The Court ultimately found no "direct control" from the U.S. and, accordingly, no infringement by the U.S. Government by *ARIEL 5*. Thus, although the Court cited *Decca* and its "master station" arguments,[84] the Court used the language, "*direct* control" as opposed to simply "control." Whether or not "direct control" is equivalent to *Decca*'s "master station" language is debatable. But, the key point in *Hughes* is that the Court essentially dropped two of the prongs in the *Decca* formula (i.e., the Court did not consider the ownership or beneficial use issues) for what constitutes "within the United States" and simply analyzed whether or not there existed "direct control."

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[82] *Id.* at 232

[83] *Id.* at 242.

[84] *Id.*

Based on the legislative history and judicial interpretation in *Hughes*, it is a safe bet that any future infringement claim against the U.S. based on the making, using, or selling of a patented invention in outer space will be invoke consideration of the extra-territorial reach of the U.S. Courts under the provisions of 35 USC § 105.

V. SPECIFIC EXEMPTIONS TO TERRITORIAL REACH-THE TEMPORARY PRESENCE DOCTRINE

Recall that 35 U.S.C. § 271(a) (the core of the U.S. Patent Act's statement on infringement) begins with the language "[e]xcept as otherwise provided in this title...." The Temporary Presence Doctrine is one such exception to patent infringement and is based on a principle that U.S. patent law does not apply to foreign flag vessels (e.g., an apparatus designed as a means of transportation or specifically, from a historical perspective, a sailing vessel) during temporary stays in the U.S. This doctrine can be traced back to the U.S. Supreme Court decision of *Brown*[85] and has found later support in the International Convention for the Protection of Industrial Property, to which the U.S. is a party. The doctrine is now codified at 35 U.S.C. § 272.[86]

As given in § 272, the Temporary Presence Doctrine is limited to "use" of an invention;[87] requires that the state of registration must afford a similar privilege to U.S.-flag vessels, aircraft, or vehicles; requires a temporary presence in the U.S.; and requires that the invention is used exclusively for the needs of a vessel, aircraft, or vehicle. For outer space activities and specifically in regard to 35 U.S.C. § 272, a special definition for the

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[85] *See Brown*, *supra* note 27.

[86] "The use of any invention in any vessel, aircraft or vehicle of any country which affords similar privileges to vessels, aircraft or vehicles of the United States, entering the United States temporarily or accidentally, shall not constitute infringement of any patent, if the invention is used exclusively for the needs of the vessel, aircraft or vehicle and is not offered for sale or sold in or used for the manufacture of anything to be sold in or exported from the United States." 35 U.S.C. § 272.

[87] The *Hughes* court stated that launching a spacecraft (i.e., a satellite), containing an embodiment of a patented invention, in a U.S. launch vehicle from U.S. territory "by or for" the U.S. Federal Government constitutes "use" of that invention within the U.S. (*Hughes*, *supra* note 81).

word "vehicle" was added in 1981 to Title III, section 305 of the National Aeronautics and Space Act of 1958 [hereinafter Space Act], the implementing legislation that created NASA.[88] Specifically, section 305(k)[89] of the Space Act states that "[a]ny object intended for launch, launched, or assembled in outer space shall be considered a vehicle for the purpose of section 272 of title 35, United States Code." This language greatly expands the definition of what constitutes a "vehicle" for purposes of 35 U.S.C. § 272 in reference to outer space activities. For example, the subject language would render cargo "launched ... in outer space" aboard the *Space Shuttle* (which is a classic example of what is commonly understood to be a space vehicle)[90] as also being a "vehicle."[91] Thus, an experimental kit, which uses a patented invention, launched by NASA onboard the *Space Shuttle* into outer space is arguably eligible for an infringement defense under 35 U.S.C. § 272.

Article 21 of the IGA implements a modified version of the traditional "Temporary Presence Doctrine" codified in the U.S. at 35 U.S.C. § 272. Article 21 states "[t]he temporary presence in the territory of a Partner State of any articles, including the *components* of a flight *element*, in transit between any place on Earth and any flight element of the Space Station registered by another Partner State or ESA shall not in itself form the basis for any proceedings in the first Partner State for patent infringement."[92] (emphasis added) Recall that 35 U.S.C. § 105(a) permits an exception to its expansion of territorial jurisdiction, i.e., what is "within the United States," by negotiation of an international agreement to which the United States is a party.[93] Because the IGA represents an international agreement to

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[88] National Aeronautics and Space Act of 1958, Pub. L. No. 85-568, 72 Stat. 426 (1958) (codified as amended at 42 U.S.C. § 2451 et seq. (2000)) [hereinafter Space Act].

[89] Now codified at 42 U.S.C. § 2457(k).

[90] Consider the language, "Any object .... launched in outer space ... shall be considered a vehicle for the purposes of [35 U.S.C. § 272]."

[91] For an interesting discussion on the expansion of the definition of a "vehicle" vis-á-vis "cargo" in light of 42 U.S.C. § 2457(k), *see Hughes*, *supra* note 81.

[92] IGA, *supra* note 16, at art. 21.6.

[93] *See* 35 U.S.C. § 105(a) (" ... except with respect to any space object or component thereof that is ... otherwise provided for by an international agreement to which the United States is a party ...").

which the United States is a party, the IGA can effectively define what constitutes U.S. territory for the purposes of U.S. patent law related to the making, using, or selling of a patented invention.

Both the Temporary Presence Doctrine defined in the IGA and the combination of 35 U.S.C. § 272 with section 305(k) of the Space Act arguably apply to cargo and are not limited to a "vessel, aircraft, or vehicle" transporting such cargo. However, the Temporary Presence Doctrine in the IGA differs from § 272 on one substantive issue. Specifically, the IGA's Temporary Presence Doctrine is not limited to an invention's "use"[94] for the exclusive needs of a vessel, aircraft, or vehicle. Thus, in accordance with the IGA, the mere transit (i.e., importation) of "any article, including components of a flight element" bound for the *ISS* through a Partner State on Earth or through a module registered by a Partner State would form the basis of a temporary presence defense against a possible infringement action. One can argue that the IGA's Temporary Presence Doctrine is a defense to 35 U.S.C. § 271(a) via 35 U.S.C. § 105 based on the national jurisdictional principle expressed in the Outer Space Treaty as well as certain language in the legislative history of 35 U.S.C. § 105.[95] However, one can counter that the statutory activities defined in 35 U.S.C. § 105 are limited to the making, using, or selling of a patented invention and do not encompass the activity of "importation" into the U.S. Thus, this line of reasoning would conclude that the mere act of importing does not activate the provisions in 35 U.S.C. § 105, and consequently, the IGA's Temporary Presence Doctrine is not available as a defense to a 35 U.S.C. § 271(a) infringement action.

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[94] Compare the language found in 35 U.S.C. § 272, which is limited strictly to "use" of the invention, leaving open the question as to whether an invention, transported through the United States for delivery to Canada and thus "imported" into the United States, is also eligible for a temporary presence defense under 35 U.S.C. § 272.

[95] "Specifically, activities occurring on space objects under the 'jurisdiction and control' of the United States are considered to have occurred in the United States; unless one of the two following exceptions [defined in 35 U.S.C. § 105] apply." S. REP. NO. 101-266, 101st Cong. (1990), at 6.

VI. DESCRIPTION AND ANALYSIS OF SOME HYPOTHETICAL CASES CONCERNING PATENT RIGHTS IN SPACE

The following hypothetical cases have been prepared to examine and discuss the current state of extra-territorial reach of the courts to subject matter concerning patents:

Example 1. Consider the scenario wherein an Italian company manufactures a device in Italy and ships it to Kazakhstan for launch aboard a Russian *Progress* supply spacecraft. The device is not a registered space object because it is considered a flight element or more specifically, cargo. The device is bound for the *ISS* and will be exclusively used and controlled from an ESA module for the benefit of the Italian company. The claims of a valid U.S. patent "read on" the device. The *Progress* spacecraft docks with the *ISS* on a Russian module and is ultimately transferred to an ESA module by transporting it through the *U.S. Lab*. May the patentee sustain an infringement action in the U.S. courts against the Italian company by virtue of its transport through the *U.S. Lab*, based on the theory that the Italian device has been imported "into the United States"?

The answer is probably not. Under 35 U.S.C. §105, the invention must be "made, used or sold in outer space on a space object...under the jurisdiction or control of the United States" to be considered to be "made, used, or sold within the United States." If the Italian device is only being transported through the *U.S. Lab*, the action of the Italian company cannot be considered to be a making, a use, nor a selling of the invention under 35 U.S.C. § 105 necessary to support a jurisdictional claim. Further, although article 21, Paragraph 2 of the IGA would seemingly expand U.S. territorial jurisdiction to any activity occurring within the *U.S. Lab* by providing that "an activity occurring in or on a Space Station flight element shall be deemed to have occurred only in the territory of the Partner State of that element's registry, except that for ESA-registered elements any European Partner State may deem the activity to have occurred within its territory,"[96] the IGA's temporary presence ex-

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[96] *See* IGA, *supra* note 16, at art. 21.2.

ception found in paragraph 6[97] would, in all likelihood, prevent a U.S. Patentee from successfully arguing an infringement based on an "importation" claim, given the activity at issue is a mere transit of the device to the ESA module. Moreover, because the Italian device will be exclusively used and controlled from the ESA module (i.e., not from U.S. territory) for the benefit of the Italian company, a U.S court is unlikely to consider the subject matter of this claim, nor the Italian company, to be of sufficient jurisdictional interest under any other theory such as one based on national jurisdiction or even one based on minimal contacts.

Example 2. Consider a patented process to capture solar energy wherein all claimed steps of the method are performed on a solar collection station on the Moon's surface. The collection station is an unregistered stand-alone unit and is not integrated into a registered space object. The process is patented in the U.S. and the rights to the patent have been assigned to a small start-up company from India that has not licensed the invention. The process is being controlled from and supplies power to a U.S. registered space object (e.g., a lunar habitat module). The solar collection station is connected to the lunar habitat module via cables. Further, the collection station is owned and operated by a commercial U.S. entity. Given that land on the Moon is *res communis* and the process is not performed on a U.S. registered space object, does the use of the patented process on the Moon's surface constitute an infringement in accordance with U.S. patent law?

In this example, the first question one should ask is whether or not the patented process associated with the collection station is "made, used, or sold in outer space on a space object or component thereof under the jurisdiction or control of the United States" under 35 U.S.C. §105(a). Because the collection station is itself an unregistered stand-alone unit, it must be determined whether the collection station is still considered to be

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[97] *Id.* at art. 21.6 ("The temporary presence in the territory of a Partner State of any articles, including the components of a flight element, in transit between any place on Earth and any flight element of the Space Station registered by another Partner State or ESA shall not in itself form the basis for any proceedings in the first Partner State for patent infringement.").

a "component" of the lunar habitat module that is registered. The analysis is further complicated by the lack of integration between the two structures. Would a court consider how integral (from an operational perspective) the collection station is to the habitat module? In this hypothetical, let us assume that the collection station is an important component for supplying critical power to the habitat module. Under this assumption, a court may conclude that a solar collection station would be a component of a space object (i.e., the habitat module) based on the close operational nexus between the collection station and the habitat module. Conversely, a drilling device for collecting lunar regolith samples (for research purposes) would *not* constitute a "component" of the habitat module under the same logic. Assuming *arguendo* that the collection station is a "component" of the U.S.-registered habitat module, then the patented process is "used in outer space" and would constitute U.S. territorial subject matter in accordance with 35 U.S.C. §105(a). Conversely, if the collection station is not considered a component of the habitat module, the next issue to address is whether or not the collection station, being owned and operated by a U.S corporation, is otherwise under the jurisdictional authority of the United States. Because the collection station is operating on the Moon, a territory of *res communis*, it arguably is not under the jurisdictional authority of the United States under the territorial theory of *Brown* or *Deepsouth*, but could perhaps fall under the jurisdictional authority of the U.S. if the logic of *Gardiner* is followed. Further, even if "control" is interpreted to mean "operational" control, the collection station is not being operated by the U.S. Federal Government in this example. Thus, the guidance provided in the other extra-territorial cases discussed *supra* of *Decca, NTP,* and *Hughes* should also be considered. Applying *NTP*, because the patented invention is a process performed entirely on the Moon, there would likely be no infringement if the collection station is not determined to constitute U.S. territory. *Decca* and *Hughes* did not distinguish between apparatus (i.e., system) and method claims. Hence, if a court applies either (a) the three-prong test of *Decca* and finds U.S. ownership, control from the U.S., and beneficial use within the United States or (b) the direct control test of *Hughes* and

finds such direct control point in the U.S., a court could find infringement.[98]

Example 3. Consider the same facts as in Example 2 with the exception that the patent claims read on the solar collection station (i.e., system claims). If the collection station is not determined to be a "component thereof" of a space object and not otherwise under the jurisdiction and control of the United States (i.e., not U.S. territory under 35 U.S.C. § 105), the situation becomes an extra-territorial scenario. First, assuming 35 U.S.C. § 105 does not apply to the collection station, 35 U.S.C. § 105 clearly applies to any invention made, used, or sold on the U.S. registered habitat module, wherein control and beneficial use takes place. Because the habitat module is a U.S.-registered space object, it in effect becomes U.S. territory under 35 U.S.C. § 105(a). Arguably, there would likely be an infringement due to the existence of U.S. ownership, control from the U.S., and beneficial use within the U.S. In other words, even if the collection station is not a "component thereof" of a U.S. space object and there is no jurisdiction or control of it by the United States, the collection station is nevertheless owned by a U.S. entity as well as controlled from and provides power to a U.S. registered space module, which *is* U.S. territory under 35 U.S.C. § 105(a) in that the invention is used by such module. The test in *Decca* requires all three of the above factors of ownership, control, and beneficial use. The test in *Hughes* requires "direct control" from the U.S. And the test in *NTP* requires control from the U.S. and beneficial use in the U.S. In sum, all tests seemingly would result in a determination of infringement under the facts of this example.

Example 4. Consider the scenario wherein a Japanese company manufactures a satellite scheduled to *commercially* launch from the U.S. (i.e., there is no international agreement to which the United States is a party). A U.S. patent's claims read on the satellite's communications subsystem and the U.S. Pat-

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[98] The Court in *NTP* was the U.S. Court of Appeals for the Federal Circuit [hereinafter Federal Circuit]. Pursuant to 28 U.S.C. § 1295, the Federal Circuit has appellate jurisdiction over patent cases. Thus, arguably, the *NTP* tests would more likely be applied in a 35 U.S.C. § 271 patent infringement case.

entee has not licensed the invention to the Japanese company. The satellite and associated launch vehicle is registered as a Japanese space object in accordance with the Registration Convention. If the satellite is processed in the U.S.; launched from the U.S.; and even directly controlled from the U.S., does the use of the satellite constitute an infringement in accordance with U.S. patent law?

35 U.S.C. § 105(a) contains two exceptions. The first exception addresses international agreements. The second exception addresses registration under the Registration Convention. Because there is no international agreement to which the United States is a party, the first exception does not apply. However, because the satellite is registered as a Japanese space object, the second exception does apply. Therefore, under this scenario, even though the satellite is controlled from the U.S. as well as was launched from the U.S., the satellite would not constitute U.S. territory under the second exception defined in 35 U.S.C. § 105(a). Further, the Japanese company may argue a temporary presence defense under 35 U.S.C. § 272. The satellite is considered a "vehicle" within the meaning of 35 U.S.C. § 272 through section 305(k) of the Space Act. And, launching a satellite in a U.S. launch vehicle from U.S. territory may be interpreted by a court as constituting "use" within the U.S. in accordance with the "use" requirement of 35 U.S.C. § 272. However even if the second exception in 35 U.S.C. § 105(a) or the Temporary Presence Doctrine of 35 U.S.C. § 272 is applicable, extra-territorial rules and guidelines may also apply. The fact that the satellite is directly controlled from the U.S. could be argued by the Patentee that the temporary presence defense is inapplicable. Further, even if the satellite itself is determined to be Japanese territory, it is directly controlled from the U.S. And "direct control" was arguably a determinative factor under the *Hughes* decision. On the other hand, both *Decca* and *NTP* require more than "control" from the U.S. Thus, whether or not infringement exists in this scenario would depend largely on which court's holding is relied on for an extra-territorial determination in the event that the exception of 35 U.S.C. § 105 or the Temporary Presence Doctrine is not considered to control the disposition.

As highlighted by the case studies above, in light of technological advancements associated with extra-terrestrial exploration and commercialization, various fact patterns may emerge in the near-term that will likely have impacts on existing patent laws. Specifically, the combination of the *res communis* principle as applied to the Moon; the lack of a significant geographical separation of nations associated with registered space modules in close proximity to each other; and some ambiguous statutory language in regard to 35 U.S.C. § 105 increases the likelihood of legal uncertainty and future litigation.

VII. POSSIBLE MITIGATION TECHNIQUES

Given the existing legal landscape as applied to patent coverage in outer space, various mitigation techniques to avoid jurisdictional issues may be of particular interest. Mitigation techniques exist for both an inventor or patent owner as well as a potential infringer. Both perspectives will be addressed.

In terms of an inventor or owner of patent rights, if an invention has a commercial application in outer space, the inventor should first consider obtaining patent coverage in all existing and foreseeable space-faring nations. In doing so, the inventor can avoid the territorial complexities associated with the issues presented in this article. However, obtaining broad patent coverage in many different nations is a complicated, time-consuming, and expensive endeavor. Further, technological advancements may result in a commercial environment whereby launches and UN registration can occur from and in practically any nation on Earth. Thus, if an inventor decides to forego broad coverage and opts to obtain limited patent coverage, another possible mitigation technique is related to claim construction. For example, in regard to U.S. patent law, given the decision in *NTP* as it relates to method claims,[99] it would be beneficial for an inventor to emphasize apparatus (i.e., system) claims in a prospective patent application as opposed to just pursuing method claims. In other words, based on the decision in *NTP*,

[99] Recall that the *NTP* Court instituted a separate test for method claims. In a method claim, every step of the method must occur within the territory of the U.S.

apparatus claims have extra-territorial reach relative to infringement actions as compared to method claims, which do not. It is noted that emphasizing apparatus claims is a common practice in patent claim drafting because in many cases, apparatus claims are considered to have greater value than comparable method claims, in part because infringement acts associated with method claims are generally more difficult to identify and enforce.

In terms of a potential infringer, it would behoove one to perform an international search of patents utilizing the known elements and limitations of the potentially infringing device as search terms. The identification of national patent coverage may help one make decisions associated with where to manufacture, launch, and possibly register a potential infringing device. Further, an infringement analysis by a competent attorney may help in identifying differences between the patented device and the potentially infringing one to permit a "design around" as well as potential risks, which can be used by a potential infringer to make informed business decisions. Such business decisions may include obtaining a license from the one or more patent owners of concern.

VIII. CONCLUSION

The environment associated with outer space activities is changing. At the time of publication of this article, governmental activities arguably dominate outer space endeavors, especially in the context of a human presence in outer space. However, it is entirely foreseeable that commercial projects in outer space will greatly increase in the near term, particularly in regard to activities in low-earth orbit [hereinafter LEO]. In fact, a human presence in LEO financed with private funds is likely in the next decade. Further, it is probable that commercial research and development activities on-board the *ISS* will dramatically increase and could form the foundation for funding the U.S. portion in regards to maintenance and operational costs of the *ISS*. The *ISS* will continue to provide an international platform for human exploration of outer space that should continue on to a return to the Moon. However, the environment

for outer space activities may change from being led by actors that are predominantly governmental to private commercial entities.

In addition to this paradigm shift, the relative proximity of space objects, registered under different nations, has the potential to strain existing patent law and the territorial nature to which it is based upon. With particular reference to U.S. patent law, lingering questions on the subject of patent infringement remain due to the intricate political web of international treaties and agreements, the use of certain ambiguous terms contained in 35 U.S.C. § 105, and the existence of judicial restraint in determining what constitutes subject matter "within the United States" for technologies that cross territorial borders.

The courts will have additional opportunities to tell us how patent infringement issues for future activities in outer space will be resolved. If history is any indication, it would appear that courts will continue to address the jurisdictional issue of extra-territorial reach as a mixed question of policy, fact, and law to be determined on a case-by-case basis.

SPACE ACT AGREEMENTS: A PRACTITIONER'S GUIDE

David S. Schuman[*]

I. INTRODUCTION

At NASA we have the best clients in the world, all with impossibly difficult jobs.[1] Depending on whom you ask, NASA's mission is to return human beings to the Moon, explore Mars, complete the *International Space Station*, determine whether life exists elsewhere, investigate the forces affecting global climate change, and characterize the size, structure, and ultimate fate of the universe. For lawyers serving the highly talented scientists, engineers, and managers working on these non-trivial problems within this great Agency, it is easy to say what the rules are, and what they cannot do. Statutes, regulations, and directives are easy to find. But the best lawyers see the bigger picture, recognize what the client is trying to do, and figure out a flexible and creative path to get there. We don't say just what the law is, we try to figure out how we can use the legal tools available to help our clients achieve their goals.

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[1] While it certainly doesn't happen every day, the author traveled to Stockholm, Sweden, in December 2006, to watch one of his clients present the Nobel Prize lecture, following the announcement he was the co-recipient of the 2006 Nobel Prize in Physics for his work on the *Cosmic Background Explorer* (*COBE*) satellite, developed at NASA's Goddard Space Flight Center. *COBE* measured the blackbody form and anisotropy (hot and cold spots) of the cosmic microwave background radiation, evidence leftover from the formation of the universe. Noted physicist Stephen Hawking, once called the result "the greatest discovery of the century, if not of all time." *See* JOHN C. MATHER & JOHN BOSLOUGH, THE VERY FIRST LIGHT 253 (1996). Dr. Mather's 2006 banquet speech contains a beautiful and succinct version of our attempt to understand the universe through the study of light. *See* John C. Mather, Nobel Prize winner in Physics, 2006, Banquet Speech (Dec. 10, 2006), http://nobelprize.org/nobel_prizes/physics/laureates/2006/mather-speech_en.html.

There's no better tool, no more flexible and creative tool, than the Space Act (hereinafter the Act).[2]

Through the exhaustive research, creative energy, and dogged initiative of Paul Dembling, General Counsel of the National Advisory Committee for Aeronautics (NACA)[3] more than 50 years ago, NASA was provided with a powerful and comprehen-sive organic statute. While it has many interesting provisions,[4] some provided in later years, there is widespread appreciation that the most useful among them is NASA's "other transactions" authority (hereinafter OTA).[5] The story of how the OTA came about is a rich and fascinating one best told by the author of that language, elsewhere in this issue of the JOURNAL OF SPACE LAW.[6] It would be no exaggeration to state that since the authority was enacted, NASA lawyers have used it to help our clients achieve their mission thousands of times. We use it every day.

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[2] The National Aeronautics and Space Act of 1958, as amended, 42 U.S.C. § 2451, et seq.

[3] Later Mr. Dembling served as NASA's General Counsel and General Counsel to the General Accounting Office [now Government Accountability Office (GAO)], followed by a 25-year partnership at the law firm of Schnader, Harrison, Segal & Lewis.

[4] Some of the more important provisions state (a) that the policy of the United States is that activities in space "should be devoted to peaceful purposes for the benefit of all mankind," (b) that NASA should "seek and encourage, to the maximum extent possible, the fullest commercial use of space," (c) that the NASA Administrator "shall be appointed from civilian life," (d) that NASA must provide for the "widest practicable and appropriate dissemination of information concerning its activities," (e) that the Agency may accept "unconditional gifts," (f) that concessionaires may operate the Agency's visitor centers, (g) that NASA may provide insurance and indemnification in certain situations, and (h) most recently, that the Agency may enter into enhanced-use leases concerning its real property. 42 U.S.C.A. §§ 2451-2473 (2003 and Supp. 2008).

[5] 42 U.S.C.A. § 2473(c)(5) (2003 and Supp. 2008).

[6] Paul G. Dembling, *The National Aeronautics and Space Act of 1958: Revisited,* 34 (2) J. SPACE L. 203 (2008). Mr. Dembling once recounted for the author some of the details concerning formation of the Space Act. Several of the most interesting aspects include the perceived jurisdictional competition from the military departments, his request to Dr. Hugh Dryden, the head of NACA, that he be allowed to draft some legislation for the new agency, the advantage obtained from providing a first draft of that legislation ("Washington operates on the first draft it gets."), his review of *every* published GAO decision on the extent of agencies' authority, and his description of how the OTA came about. Telephone Interview with Paul G. Dembling (July 25, 2002) (transcript available upon request).